

Lu Mengchen

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EDUCATION BACKGROUND

Lancaster University	09/2020-06/2024
- BEng Hons Electronics and Communications Engineering Bachelor of Engineering - Second Class Honours (Division one)	
Beijing Jiaotong University	09/2020-06/2024
BEng Communication Engineering GPA:3.03/4.0 Weighted Average Mark: 80.3/100	
Nanyang Technology University(Singapore)	09/2024-01/2026
Master degree Major: Communication Engineering(IoT) 2024 Fall	

INTERNSHIP EXPERIENCE

Electronic Information Engineer Intern, Beijing ZX-CE Technology Co., Ltd.	06/2023-09/2023
- Wireless Network Planning: Engaged in comprehensive network planning based on existing network conditions, such as core network server addresses and operator information. This included hardware selection, configuration, and wireless parameter planning. - Network Troubleshooting: Conducted specific network testing after completing network planning to ensure normal operation and achieve performance standards. - Wireless Network Optimization: Analyzed and optimized the network based on actual testing data to improve network performance and service quality, achieving outdoor coverage requirements of -95dBm or higher and SINR requirements of 3dB or higher. - Participated in the mobile base station signal measurement project at Xi'an University of Posts and Telecommunications (Chang'an Campus) over a period of three months, and collaborated with teammates to optimize the signals of 56 base stations within the campus.	
Electronic Information Engineer Intern, China Mobile	12/2022-03/2023
- Participated in training on network fundamentals, analysis and testing tools, and software, developed proficiency in commonly used tools such as Probe and MapInfo through an intensive one-week training. - Joined the testing team and conducted testing and analysis on key targets, VIP complaints, and grid layout. Identified and addressed 36 problem areas and provided solutions for 42 adjustments. - Attended advanced training in signaling and backend analysis, acquired preliminary skills in event analysis and developed the capability to analyze events such as call drops and poor quality, as well as gained a basic understanding of the main backend parameters, with the ability to verify and modify data. - Joined the backend analysis team, focusing on analyzing critical scenarios and improving degraded areas. Successfully identified and resolved 226 problem areas, ensuring a closed-loop adjustment. Additionally, led the key scenario holiday support projects in New Year's Day and Spring Festival , effectively addressed 26 hidden issues.	

PROJECT EXPERIENCES

Intelligent Tracking Robot (Project Hardware Lead)	02/2023-09/2023
- Responsible for the overall structural design of an intelligent car - Ensured balance and simulated movement - Designed and updated the control PCB five times - Conducted full vehicle PID tuning	
Operational Amplifier Bridge Current Detector(Individual Project)	05/2022-11/2022
- The project aims to use LM324 operational amplifiers to design a bridge circuit for testing circuit errors. - Utilized NI Multisim to design the circuit topology model with operational amplifiers, soldered the physical circuit using the appropriate methods, and ensured its functionality. - Created a theoretical model of the breadboard using Electronic Pioneer software and validated it using the Windows platform.	
Automatic Image Processing Program Design in Matlab(Individual Project)	04/2022-08/2022
- Developed a program aimed at utilizing Background subtraction technology for detecting anomalies in images, with application in the surveillance field. - Implemented YCbCr image format conversion and utilized percentage-based image compression techniques using	

Matlab.

- Successfully completed the entire project and achieved an excellent mark of 85.

Intelligent Reverse Radar Project Leader

07/2021-10/2021

- Led a project aimed at improving the existing vehicle-mounted reverse radar by changing the buzzing frequency to volume as a reference indicator for distance.
- Developed Arduino code to establish the connection between the buzzer and the radar. Designed a distance-decibel formula using Arduino software and incorporated it into the code to increase volume with decreasing distance.
- Designed and assembled the physical components using an Arduino board, buzzer, ultrasonic sensor, LED, and soldering tools.

EXTRACURRICULAR ACTIVITIES

President, BJTU Student Union Sports Department

2020-2023

- Organized and participated in sports meetings, overseeing tasks such as venue arrangement and referee training;
- Holder of the shot put record at BJTU.
- Responsible for officiating various intra-school sports events.

Representative of Lancaster University, Europe 101 Global Leadership Meeting

08/2022-03/2023

- Represented Lancaster University in presenting the perspectives of Asian international students, dedicated to addressing global issues through global leadership and proposing solutions for the youth.

Director of Admissions Promotion Office, Beijing Jiaotong University

2020-2022

- Managed and coordinated admissions promotion activities as the leader of the admissions campaign and the head of the admissions practice team in the Jilin region.
- Represented BJTU in the "Hundred Universities Enter Baishan" special event, and participated in a special forum on this event at Jilin Provincial TV to promote our university.

AWARDS & ACHIEVEMENTS

Beijing Jiaotong University Cultural and Sports Excellence Scholarship in the 2022 academic year	08/2022
Social Work Excellence Scholarship in the 2022 Academic Year	06/2022
2021 Winter Vacation Admissions Promotion Social Practice Activity Volunteer Advanced Individual	12/2021
2021 Winter Vacation Admissions Promotion Social Practice Activity Volunteer Outstanding	12/2021
Admissions Practice Group Leader	

ADDITIONAL INFORMATION

Languages: Chinese(native), English(IELTS 7.0, Listening 7.5, Reading 8.0, Writing 6.0, Speaking 6.0)

Computer Skills: MS Office, PS, PR, AutoCAD, Inventor, EasyEDA, Matlab

Interests: Playing Badminton, Archery, Swimming